

OTT Streaming Content Trend Analysis

Exploratory Data Analysis (EDA) Report

Dataset: Netflix Titles | 8,807 titles | 12 columns

Tools: Python, pandas, NumPy, Matplotlib, Seaborn

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Role: Data Analyst | October 2026

1. Introduction and Objective

Streaming platforms compete on the breadth and freshness of their content libraries. This report applies exploratory data analysis (EDA) to the Netflix titles catalogue to understand **what the library contains, where it comes from, and how it has changed over time.**

Key questions

- How is the catalogue split between Movies and TV Shows, and is that mix changing?
- When was most of the content added to the platform?
- Which genres, countries and maturity ratings dominate?
- Have movie lengths and TV show lifespans changed?

2. Dataset Overview

The dataset has **8,807 titles** and **12 columns**. Each row is one title, identified by a unique `show_id`. There are no duplicate rows.

Column	Description	Missing
<code>show_id</code> , <code>type</code> , <code>title</code>	Unique ID, Movie / TV Show, title name	0
<code>director</code>	Director name(s)	2,634 (29.9%)
<code>cast</code>	Lead cast members	825 (9.4%)
<code>country</code>	Production country or countries	831 (9.4%)
<code>date_added</code>	Date the title was added to the platform	10 (0.1%)
<code>release_year</code>	Year the title was originally released	0
<code>rating</code>	Maturity rating (TV-MA, PG-13, ...)	4
<code>duration</code>	Minutes (movies) or seasons (TV shows)	3
<code>listed_in</code> , <code>description</code>	Genre tags and a short synopsis	0

Cardinality check: 2 content types, 15 ratings, 74 release years, 514 distinct genre combinations and 749 distinct country strings (many are multi-country combinations, which are split in Section 3).

3. Data Quality and Cleaning

Four issues were found and handled before analysis. Each fix is deliberately small and traceable.

Issue found	Evidence	Action taken
Shifted values in 3 rows	Three Louis C.K. titles had the runtime (e.g. "74 min") in the <i>rating</i> column and a null <i>duration</i>	Moved the value to <i>duration</i> , set <i>rating</i> to null
Missing country and rating	831 null countries; 4 null ratings (incl. the 3 above)	Filled with the label "Unknown" so rows are kept
Dates stored as text	<i>date_added</i> held strings like "September 25, 2021"	Parsed to datetime; derived <i>year_added</i>
Duration mixes units	"90 min" vs "2 Seasons" in one column	Extracted the number into <i>duration_num</i> ; analysed Movies and TV Shows separately
Multi-valued fields	Genres and countries are comma-separated lists	Exploded into long tables: 19,323 genre rows and 10,845 country rows

Handling of missing data

director (30% missing) and *cast* were not imputed and are not used in the analysis. Filling them would add no real information. *country* and *rating* were labelled "Unknown" and excluded from rankings so they do not distort the top-10 charts.

4. Exploratory Analysis

4.1 Content mix: Movies vs TV Shows

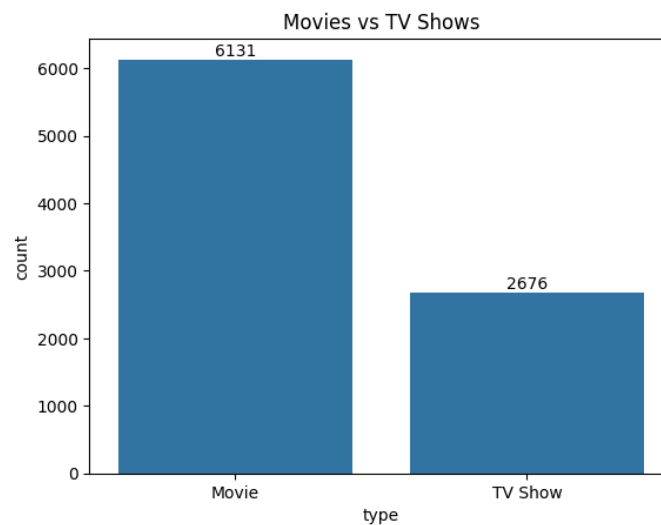


Figure 1. Count of titles by type

Movies make up **69.6%** of the catalogue and TV Shows **30.4%**, roughly a 7 : 3 split.

Insight: The library is movie-heavy overall, but Section 4.3 shows that this is changing in newer releases.

4.2 When was content added?

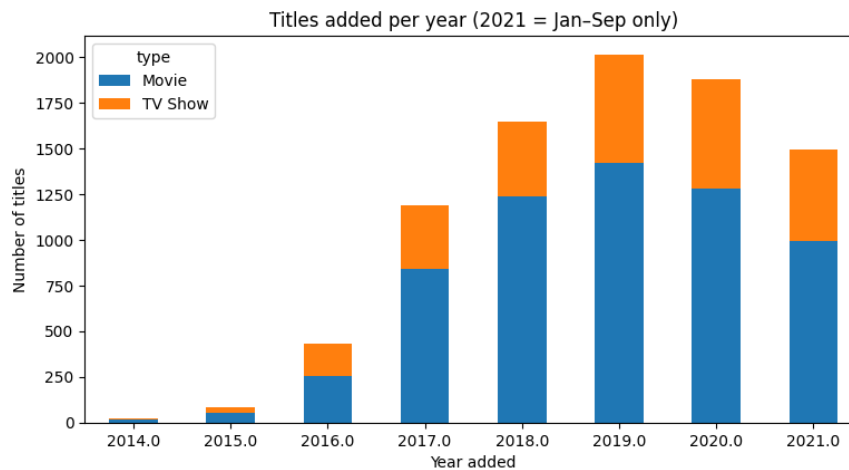


Figure 2. Titles added per year, 2014-2021 (2021 covers January-September only)

Year added	2014	2015	2016	2017	2018	2019	2020	2021*
Titles	24	82	429	1,188	1,649	2,016	1,879	1,498

*Partial year

Insight: Additions grew about 84x from 2014 to the 2019 peak of 2,016 titles. 2020 dipped slightly. 2021 is only a nine-month figure but is already on pace to match or exceed 2020, so the dip should not be read as a decline.

4.3 TV share of releases over time

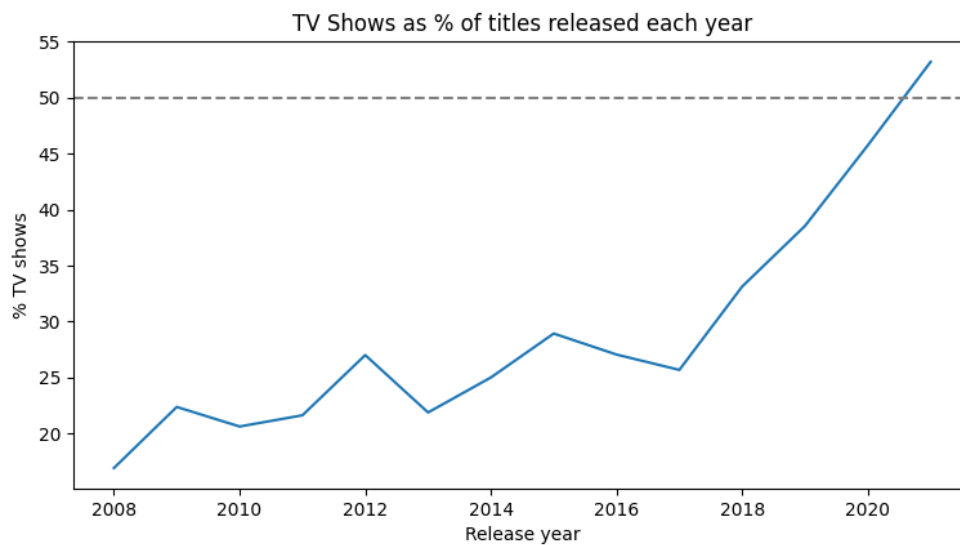


Figure 3. TV Shows as a percentage of titles released each year (2008-2021)

TV Shows were about **25%** of titles released in 2014 and about **53%** in 2021. The line crosses the 50% reference line, so TV Shows are now the majority of the most recent releases.

Insight: The overall 70/30 split reflects the back catalogue. Recent releases tell a different story: the platform is shifting toward series-based content.

4.4 Top genres

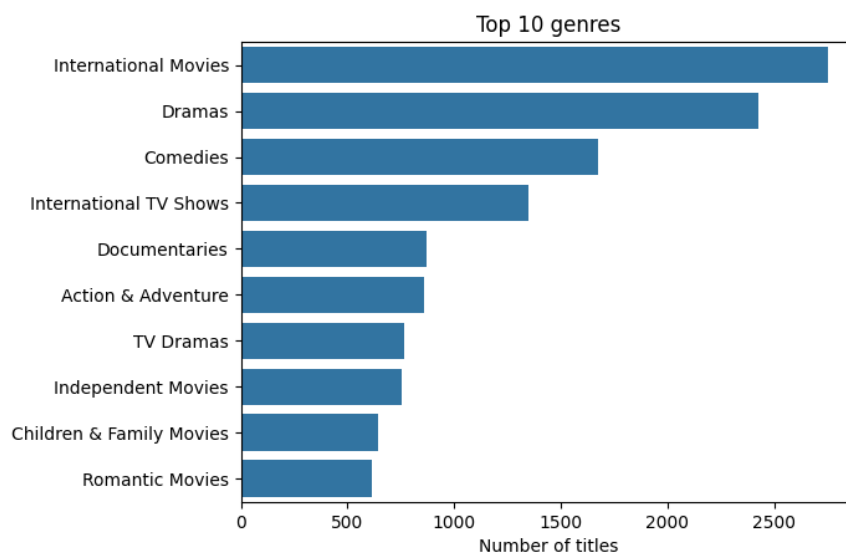


Figure 4. Top 10 genres (a title can carry several genre tags)

International Movies (about 2,700 tags) and Dramas (about 2,400) lead, followed by Comedies, International TV Shows, Documentaries and Action & Adventure.

Insight: International content sits at the top of the genre ranking in both the movie and TV categories, which signals a strongly global catalogue.

4.5 Top producing countries

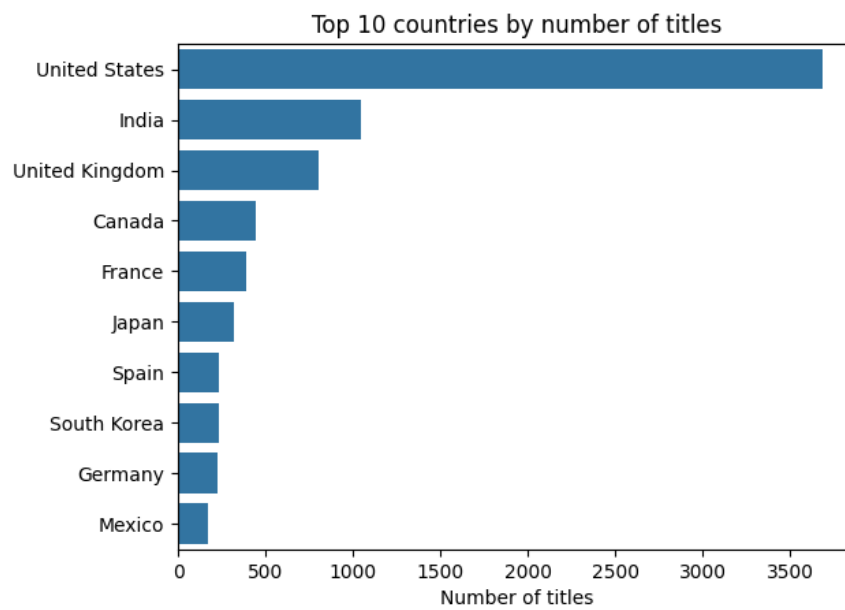


Figure 5. Top 10 countries by number of titles (Unknown excluded; co-productions count for each country)

The United States leads clearly with roughly 3,700 titles, followed by India (about 1,000) and the United Kingdom (about 800). Canada, France, Japan, Spain, South Korea, Germany and Mexico complete the top ten.

Country share by release period (% of country-title entries)

Period	United States	India	South Korea	Spain
2000-10	39.2	15.8	0.6	0.9
2011-15	32.0	12.8	2.1	1.5
2016-18	34.8	8.2	2.9	3.1
2019-21	39.5	8.2	3.1	3.2

Insight: The US share is stable at roughly a third to two-fifths. India's share has fallen from about 16% to 8%, while South Korea and Spain have each grown roughly fivefold and 3.5-fold respectively, from very small bases.

4.6 Maturity ratings

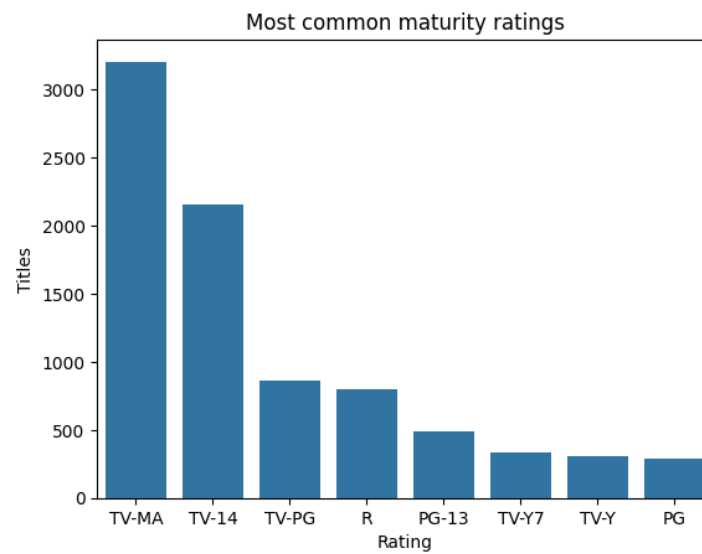


Figure 6. Most common maturity ratings

TV-MA is the most common rating, followed by TV-14. Together they account for **61%** of all titles.

Insight: The catalogue is skewed toward teen and adult audiences. Kids' ratings (TV-Y, TV-Y7, PG) are each a small share.

4.7 Movie runtime

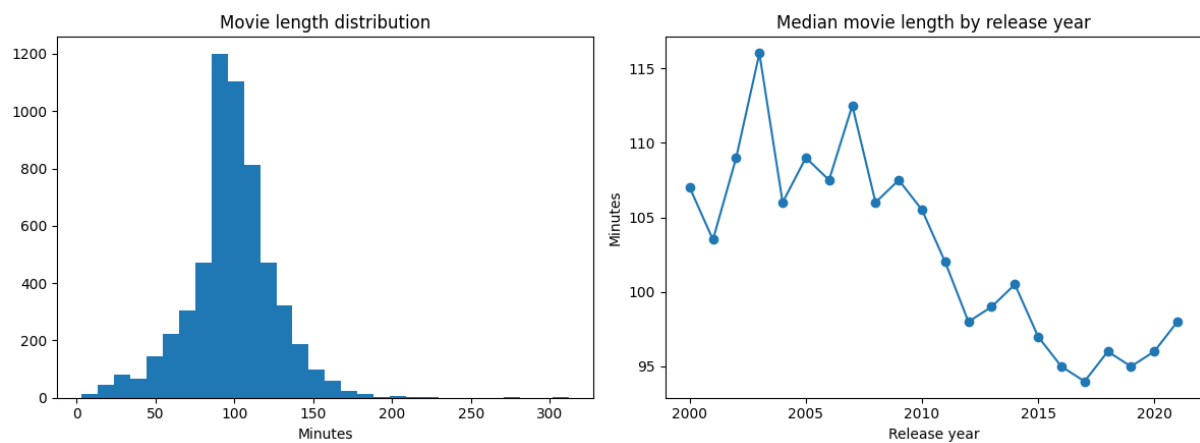


Figure 7. Movie length distribution (left) and median runtime by release year (right)

Runtimes are roughly bell-shaped and centred near 90-110 minutes, with a thin right tail of very long titles. The median runtime fell from **108 min** for titles released before 2010 to **96 min** for 2018 onward, a drop of about 12 minutes.

Insight: Newer movies on the platform are shorter. The median hovered above 105 minutes in the mid-2000s and settled in the mid-90s after about 2015.

4.8 TV shows by number of seasons

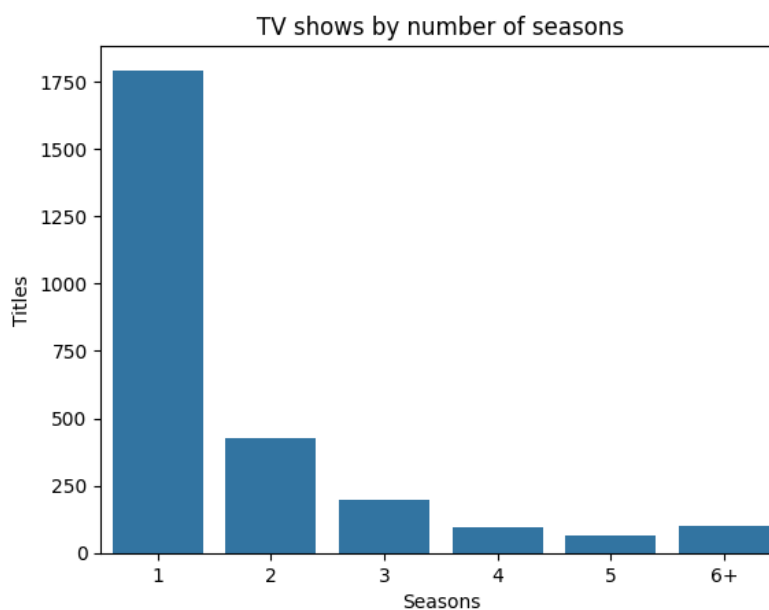


Figure 8. TV shows grouped by number of seasons (6+ combined)

67% of TV shows have just one season.

Insight: Most series on the platform are short-run or limited series. Long-running shows are rare.

5. Summary of Key Findings

Area	Finding
Content mix	70% Movies / 30% TV overall, but TV reached about 53% of 2021 releases (up from about 25% in 2014)
Growth	Titles added per year rose from 24 (2014) to a 2,016 peak (2019)
Genres	International Movies and Dramas lead; International TV Shows is the top TV genre
Geography	US dominates (about 3,700 titles); India is second; Korea and Spain are rising
Audience	TV-MA and TV-14 make up 61% of the catalogue
Length	Median movie runtime fell from 108 to 96 minutes
Series length	67% of TV shows have only one season

6. Conclusion and Recommendations

The analysis shows a catalogue that expanded rapidly between 2015 and 2019, is increasingly international, and is shifting toward shorter movies and series-based content aimed at teen and adult viewers.

Suggested next steps

- Add a **country-by-genre** view to see which regions drive the International categories.
- Compare **release year vs year added** to measure how quickly new titles reach the platform.
- Merge in viewership or rating data (e.g. IMDb) to link content trends to audience response.
- Use the ~12-minute movie runtime decline as a hypothesis for further testing.

Limitations

- The dataset is a catalogue snapshot. It contains no viewership, so "popular" cannot be inferred from it.
- 2021 covers January-September only.
- Genre and country counts are tag-based, so one title can count toward several categories.
- Director (30%) and cast (9%) have notable missing data and were not analysed.